



NAMIBIA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Introduction to Linux

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Learning Outcomes

- Understand Linux fundamentals
- Navigate the Linux filesystem
- Manage files and directories
- Use permissions and processes
- Write basic shell scripts
- Prepare for HPC usage



What is Linux

- Linux is a free, open-source operating system.
- Just like Windows or macOS, it manages hardware resources and allows software to communicate with the physical machine
- Widely used for:
 - High Performance Computing , Cloud computing, Data Science, Cybersecurity, Web Servers, Desktop & Personal Computing, etc.



Why Linux for HPC

- Stable Secure
- Open-source and stable
- Dominates TOP500 supercomputers
- Excellent performance and security
- Supports MPI, OpenMP and GPU computing
- Powerful automation through shell scripting
- Supports MPI and GPU computing
- Handles thousands of users
- Excellent scripting



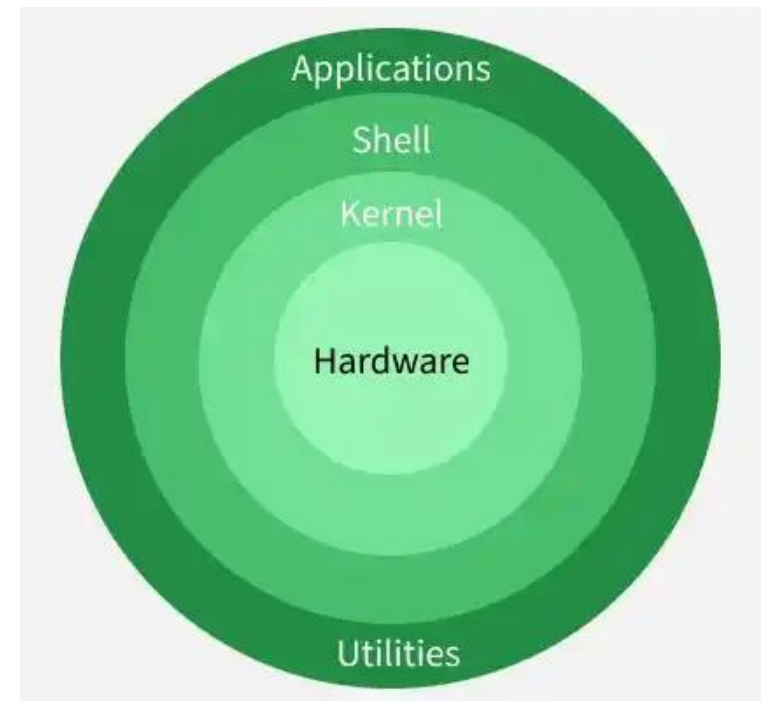
Basic Principals of Linux

- Everything is a file
- Linux is case-sensitive
- MYFILE.doc, Myfile.doc, mYfiLe.Doc are different.



Linux Architecture

- User
- Bash Shell
- Linux Kernel
- Hardware





Linux File System

- The Linux file system organizes data in an inverted tree structure, starting at a single root directory (/).
 - /home - user files
 - /etc - configuration
 - /opt - applications
 - /tmp - temporary files
 - /usr - system programs
 - /var - logs



Linux File System

- Every file in Linux can be one of the following four types:
 - Ordinary File (contain data, info)
 - Directories (hold files & other directories)
 - Devices (for accessing the hardware)
 - Links (Pointer to another file)
 - Hard Link
 - Soft Link



- Two types of login screens
 - virtual consoles (text-based)
 - graphical logins (display managers)
- Login using login name and password
- Each user has a home directory for personal file storage
- **passwd**, change password
 - **ppasswd**
- **id**, print user and group ids
- **su**, switch user, **su –** (complete user environment)
 - **su - bob**



Basic Linux Commands

- Current directory
 - `pwd`
- List files
 - `ls`
- Long format
 - `ls -l`
- Hidden files
 - `ls -a`
- Human readable
 - `ls -lh`



- **man**, manual pages
 - man <command>
 - man ls
- **--help**, basic help by author
 - ls --help



- Move into a directory
 - `cd documents`
- Go home
 - `cd`
- Previous directory
 - `cd -`
- Root
 - `cd /`
- The Root Home Directory
 - `cd /root`
 - is simply the personal home directory for the superuser (the root account)



- Navigate to
 - home
 - /
 - List all files including hidden files in a long format
 - /tmp
 - back home



- **Files**
- **touch**, create empty files or update file timestamps
 - touch file1.txt
 - touch FILE1.txt
- Multiple files
 - touch file2 file3
 - touch file{1..5}.txt
- **Directories**
 - mkdir HPC
 - mkdir files
- Nested
 - mkdir -p HPC/Lab1/Data



Copy and Move Files

- **Copy Files**
 - `cp file1 file2`
- **Copy Directories**
 - `cp -r folder backup`
- **Move Files**
 - `mv old.txt new.txt`
 - `mv report.txt Documents/`



Delete Files

- Files
 - `rm file4.txt`
 - `rm -i file5.txt`
- Directory
 - `rm -r folder`
- Never use
 - `rm -rf /`



Exercise

- Create a directory named **Workspace** and a subdirectory inside it called **Drafts**.
- Inside the Drafts folder, create three empty text files named **file1.txt**, **file2.txt**, and **file3.txt**.
- Copy **file1.txt** into the parent **Workspace** directory.
- Move **file2.txt** into the **Workspace** directory and simultaneously rename it to **important.txt**.
- Create a copy of the **Drafts** directory (including its contents) and name the copy **Backup_Drafts**.
- Remove **file3.txt** from the **Drafts** directory.
- Remove the **Backup_Drafts** directory and all the files inside



- **cp /etc/rsyslog.conf .**
- **mv rsyslog.conf file.doc**
- cat file.doc
- More file.doc
- Less file.doc
- head file.doc
- tail file.doc
- Live log
 - tail -f



- `cat` can also be used to create a new file.
 - `cat > file4`
 - The Ctrl+d key combination will send an EOF (End of File) to the running process.
- It can also be used to append text to end of an existing file with
 - `'>>'`(double greater than) symbol.
 - `cat >> lsa`



- Find file
 - `find . -name "*.txt"`
- Locate text



- Linux wildcards (also known as globbing patterns) are special symbols used by the shell to match files, directories, or text strings automatically
- **Asterisk *** - Zero or more characters.
 - `ls *.txt,`
 - `cp file.* dir2`
 - `cp *.pdf Documents/`
 - `rm *`
 - `rm *.txt`



- **Question Mark ?** - Exactly one character.
 - `rm file?.log`
 - `ls file?.txt`
 - `ls ?.txt`
 - `cp fil??.txt Desktop`
- **Square Brackets []** - A single character from a defined set/range.
 - `cp image[1-3].png backup`
 - `ls file[12].txt`
 - `ls [a-c]*`



Combined Wildcards

- `ls file[1-3]?.txt`
- `ls [A-Z]*.txt`
- `ls report[1-3]*?.txt`
- `ls data[2-5]?.*`
- `ls report[1-3][A-C].txt`
- `ls report?? [1-3][A-C].txt`



- Create the following files:
 - touch report1.txt report2.txt report3.txt report10.txt
 - touch data1.csv data2.csv data3.csv data4.csv
 - touch apple.txt banana.txt cherry.txt orange.txt
- Display **all report text files**, regardless of the number that follows the word *report*.
- Display **only the report text files that have a single-digit number** in their filename.
- Display **only** data2.csv, data3.csv, and data4.csv.
- Display **all files whose names begin with the letters a, b, or c**.
- Display **all CSV files** in the current directory.
- Display **all files whose names begin with data**, regardless of their extension.



Exercises

- `ls report*.txt`
- `ls report?.txt`
- `ls data[2-4].csv`
- `ls [a-c]*`
- `ls *.csv`
- `ls data*`



- **wc**
- – The `wc` (word count) command is used to find out the number of lines, words and characters in given file(s).
 - `wc file3`
 - `wc -l` : Prints the number of lines in a file.
 - `wc -w` : prints the number of words in a file.
 - `wc -c` : Displays the count of bytes in a file.
- **Pipe (|)**
- A pipe (|) takes the output of one command and sends it as the input to another command.
 - `ls | wc -l`



- **Grep**
- grep searches for text inside files.
 - `grep <your username> /etc/passwd`
 - `cat data | grep python`
 - `grep -i linux notes.txt`
- **cut**
- cut extracts selected columns from a file.
 - `cut -d "," -f1 employees.csv`
 - `cut -d "," -f1,4,7 /etc/passwd | tail -4`

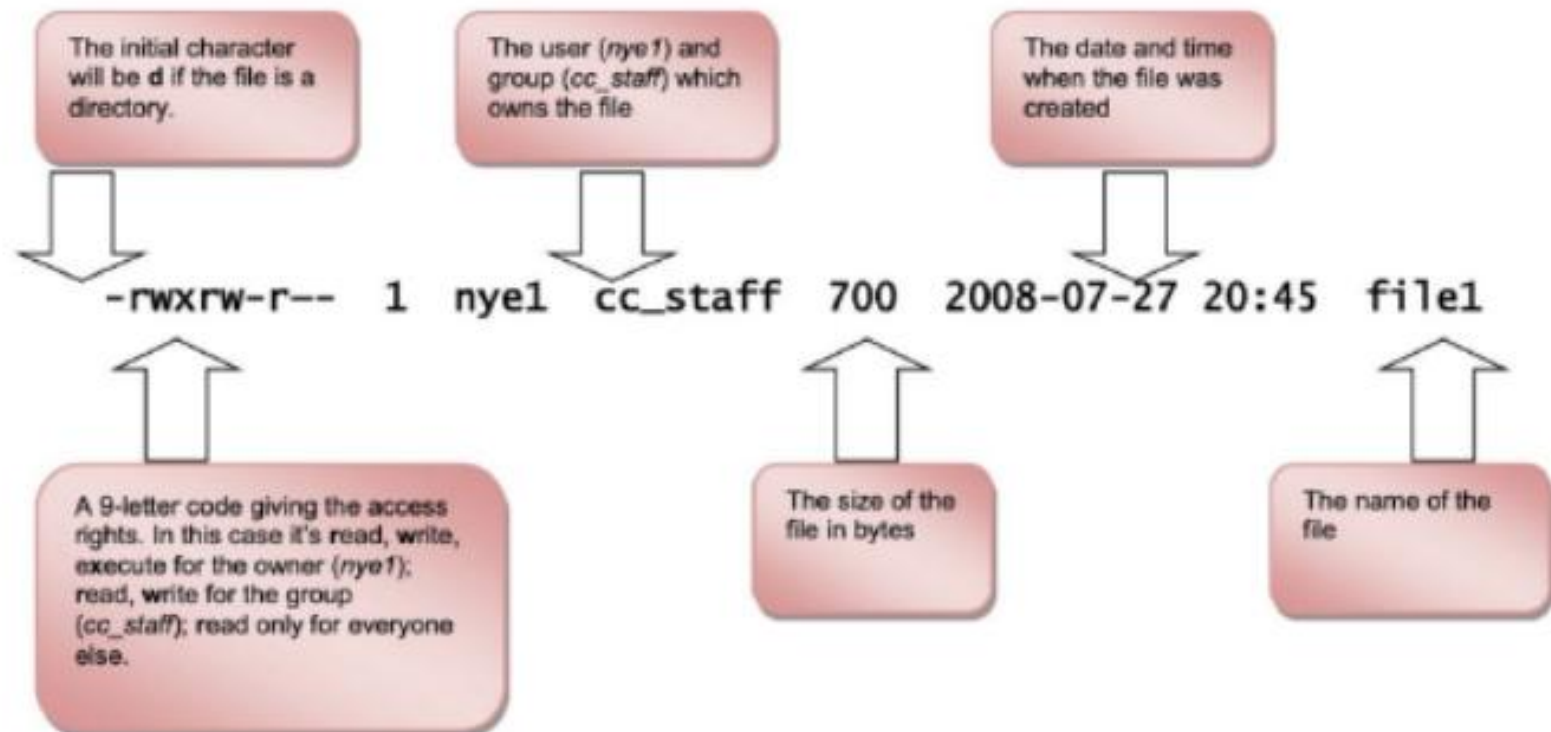


- **sort**
- Sorts text alphabetically or numerically.
- Alphabetical
 - `sort names.txt`
- Reverse
 - `sort -r names.txt`
- **uniq**
- **uniq removes adjacent duplicate lines.**
 - `uniq fruits.txt`
 - `Sort fruits.txt | uniq`
- **Count Duplicates**
 - `uniq -c fruits.txt`



Permissions

- Linux permissions control who can read, write, or execute files and directories, acting as the system's core security model.
- Viewing file and directory permissions (**ls -l**)





- Permissions are based on four access types, or rules.
 - (r) Read permission
 - (w) Write permission
 - (x) Execute permission
 - (-) No permission or no access
- These permissions can be applied to three classes of users:
 - Owner (U) The owner of the file or application
 - Group (G) The group that owns the file or application
 - Other (O) All users



- Numeric Representation of Permissions
- Linux permissions are represented using a numeric (octal) notation.

Letter	Permission	Value
r	Read	4
w	Write	2
x	Execute	1



- These values can be added together to create combinations:
 - 7 (4+2+1) = read, write, and execute
 - 6 (4+2) = read and write
 - 2 = write only
 - 1 = execute only
 - 0 = no permissions
- The permission `chmod 751` sets:
 - Owner: 7 (read, write, execute)
 - Group: 5 (read, execute)
 - Others: 1 (execute)



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 - The permission `chmod 751` sets:
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 - Others: 1 (execute)
- `chmod 535 newfile.txt`
`chmod 745 dir4`
`chmod 731 -r dir1`



Processes

- Running processes
 - ps
- Interactive
 - top
- Kill
 - kill PID
 - killall sleep
 - sleep 1000 &
 - ps



Monitor storage space

- du (Disk Usage)
 - df -h
 - du -sh
- df (Disk Free) - Directory size
 - df -h
 - df -Th



- nano
- Modeless terminal text editor. Start typing immediately after opening a file. Shortcuts are displayed at the bottom of the screen, where the caret symbol (^) denotes the Ctrl key, and M represents the Alt key,
 - nano file3.doc
 - Ctrl+X: save and quit
 - Ctrl+O: Write Out
 - Ctrl+R: Insert another file



- Overwrite:
 - `ls > files.txt`
 - `cat file3 > newfile`
- Append:
 - `echo "Hello" >> files.txt`



- **clear**, clears the screen
- **date**, see/modify system date & time
 - date, date [MMDDhhmm[[CC]YY][.ss]]
- **hostname**, show or set the system hostname
 - hostname [<new-name>]
- **uname**, print system info
 - uname



- **scp**, secure copy
- Copy a File from Local Computer to the HPC Cluster
 - `scp <file name> <username>@<hostname>:<path>`
 - ***scp C:\Users\Bob\Documents\data.csv shoopala@10.100.0.100:/home/bob***
- Copy a Directory from the HPC Cluster to the Local Computer
 - `scp <username>@<hostname>:<path> <local path>`
 - ***scp bob@10.100.0.100:/home/bob/output.csv C:\Users\Bob\Downloads***



- The tar (Tape Archive) command in Linux is used to bundle multiple files and directories into a single archive file.
- Create
 - `tar -czf data.tar.gz Data`
- Extract
 - `tar -xzf data.tar.gz`
- Using gzip compression with tar
- Create
 - `tar -cvzf file.tar.gz *.c`
- Extract
 - `tar -xvzf file.tar.gz`



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Questions